Lecture Notes in Computer Science 15731

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Cinzia Di Giusto · António Ravara
Editors

Coordination Models and Languages

27th IFIP WG 6.1 International Conference, COORDINATION 2025
Held as Part of the 20th International Federated Conference on Distributed Computing Techniques, DisCoTec 2025
Lille, France, June 17–19, 2025, Proceedings

Springer

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ISSN 0302-9743 ISSN 1611-3349 (electronic)
Lecture Notes in Computer Science
ISBN 978-3-031-95588-4 ISBN 978-3-031-95589-1 (eBook)
https://doi.org/10.1007/978-3-031-95589-1

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Foreword

The 20th International Federated Conference on Distributed Computing Techniques (DisCoTec 2025) took place from 16th to 20th of June, 2025 at the Lille campus of *Arts et Métiers* (ENSAM) and was organised by the Research Center in Computer Science, Signal and Automatic Control of Lille (CRIStAL) with support from the University of Lille, the Inria Centre at the University of Lille, and the French National Centre for Scientific Research (CNRS).

The DisCoTec series is one of the major events sponsored by the International Federation for Information Processing (IFIP). It comprises three conferences:

- The IFIP WG6.1 International Conference on Coordination Models and Languages (COORDINATION)
- The IFIP WG6.1 International Conference on Distributed Applications and Interoperable Systems (DAIS)
- The IFIP WG6.1 International Conference on Formal Techniques for Distributed Objects, Components, and Systems (FORTE)

Together, these conferences cover a broad spectrum of distributed computing subjects, ranging from theoretical foundations and formal description techniques to systems research issues. Following the established tradition and in addition to the individual sessions of each conference, the event comprised several plenary sessions gathering attendants from the three conferences. These included four joint invited speaker sessions, a joint session for the best papers from each of the three conferences, a session dedicated to gender parity and the role of women in science, an event dedicated to young researchers, and a session dedicated to demonstration of the artefacts created by the community.

The three main conferences were further complemented by five satellite events:

- the workshop "Components Operationally: Reversibility and System Engineering" (CORSE), in honour of Jean-Bernard Stefani on the occasion of his 65th birthday
- the 18th International Workshop on Interaction and Concurrency Experience (ICE 2025)
- the 1st International Workshop on Adaptable Cloud Architectures (WACA 2025)
- three invited tutorials—one from each main conference—showcasing outstanding tools of interest to the community
- a full-day tutorial presenting the ΔQSD design paradigm

I would like to thank the Program Committee chairs, the Artefact Evaluation chairs, and the organisers of all the events for their involvement and cooperation during the preparation of the conference, and the Steering Committee and Advisory Boards of DisCoTec and the three main conferences for their guidance and support. The organisation of DisCoTec 2025 was only possible thanks to the dedicated work of the Organising Committee, including Bas van den Heuvel and Matthew Alan Le Brun (Publicity chairs), Roberto Casadei (Artefact Evaluation coordinating co-chair), Larisa Safina (Satellite

Events chair), Imen Sayar and Manel Barkallah (Gender Parity / Women in Science Co-chairs), Adrien Luxey-Bitri (Young Researchers Forum chair), Alexandre Desquesne, Aissa Ainaoui, and Isabelle Aslani (logistics and finances), as well as all the students who volunteered their time to help, particularly Nathan Leblond and Rémy Raes (Head of Student Volunteers). Many thanks to Arts et Métiers, and particularly to Rodolphe Sory, for providing the infrastructure to host the event. Finally, I would like to thank IFIP WG6.1, Springer's Lecture Notes in Computer Science team, the University of Lille, the Inria Centre at the University of Lille, and the CRIStAL laboratory for their support and sponsorship. EasyChair provided the reviewing platform, whereas CNRS has provided the registration and accounting infrastructure. Last but not least, thanks to the DisCoTec community—both speakers and attendees—for making it happen.

Happy 20th anniversary, DisCoTec!

June 2025 Simon Bliudze

Preface

This volume contains the papers presented at COORDINATION 2025: the 27th International Conference on Coordination Models and Languages, held on June 17–19, 2025, in Lille.

Modern information systems rely increasingly on combining concurrent, distributed, mobile, adaptive, reconfigurable, and heterogeneous components. New models, architectures, languages, and verification techniques are necessary to cope with the complexity induced by the demands of today's software development. Coordination languages have emerged as a successful approach, in that they provide abstractions that cleanly separate behaviour from communication, thereby increasing modularity, simplifying reasoning, and ultimately enhancing software development. COORDINATION provides a well-established forum for the community of researchers interested in models, languages, architectures, and implementation techniques for coordination.

COORDINATION 2025 solicited contributions in three different categories: (1) regular papers describing thorough and complete research results and experience reports; (2) tool papers describing technological artefacts in the scope of the research topics of Coordination; and (3) survey papers describing important results and success stories that originated in the context of coordination. Artefacts submitted with papers were also evaluated by three members of the Artefact Evaluation Committee, which awarded EAPLS badges for being Available, Functional, Reusable, and with results Reproducible.

There were 27 submissions. Three program committee members single-blindly reviewed each submission. The committee decided to accept 12 papers, 10 regular and 2 tool papers. The program also included 1 invited talk and 1 invited tutorial.

Four accepted papers are on choreographic approaches to model and validate concurrent and distributed systems. Christopher Esterhuyse, Benjamin Lion, Hans-Dieter Hiep, and Farhad Arbab authored the paper "Formal Foundations for Reowolf: Multi-Party Sessions via Synchronous Protocol Programming", awarded two badges (available and functional), which defines formally the semantics of Reowolf connectors to compose message-passing multiparty applications defined with the synchronous protocol description language PDL. The correctness criteria of PDL runtime systems and the connector runtime reference implementation were machine-checked. Robert Rubbens, Petra van den Bos, and Marieke Huisman authored the paper "Verified Parameterized Choreographies", awarded two badges (available and functional), which presents an extension of the choreographic verification platform VeyMont with parameterized endpoint families and parameterized communication. The semantics of the language is defined by a choreographic projection, which encodes a parameterized choreography into a single PVL program that the platform VerCors can verify. This new version of VeyMont also generates provably correct code. Nikolaus Huber, Susanne Graf, Philipp Rümmer, and Wang Yi authored the paper "MIMOSA: A Language for Asynchronous Implementation of Embedded Systems Software", awarded two badges (available and reusable), which

presents the Mimosa language for the design and implementation of asynchronous reactive systems communicating through FIFO buffers. The semantics supports side-effectful computations and has an asynchronous coordination layer to orchestrate processes. Interested readers can experiment with an interpreter. Tilman Zuckmantel, Thomas Hildebrandt, Yongluan Zhou, and Boris Düdder authored the paper "DACEO: Declarative Asynchronous Choreographies with Data-dependent Event Ordering", which presents a formal modelling language designed for declarative asynchronous choreographies with data-dependent message ordering and data objects which enforce causal consistency constraints in an asynchronous, event-based micro-services architecture.

On the orchestrator-based view of modelling distributed systems, there are two papers. Giorgio Delzanno, Cosimo Laneve, Arnaud Sangnier, and Gianluigi Zavattaro authored the paper "Decidability Problems for Micro-Stipula", which studies the borders between decidability and undecidability of determining whether a clause is never executed in Micro-Stipula, a language to describe smart contracts. Carlos Gustavo Lopez Pombo, Agustín Eloy Martinez-Suñé, Hernan Melgratti, Diego Senarruzza Anabia, and Emilio Tuosto authored the paper "Behavioural, Functional, and Non-Functional Contracts for Dynamic Selection of Services", which discusses bisimulation techniques for automata models for expressing service contracts and checks that the interoperability can be guaranteed even when contracts use mismatching names.

Two Tool papers are included in these proceedings. José Proença and Maurice H. ter Beek present "RebeCaos", awarded two badges (available and functional), which describes a web-based front-end tool for the Rebeca language. Rebeca is an actor-based language for modelling and analysing concurrent and distributed systems. RebeCaos can simulate different operational semantics of Rebeca, together with some reachability analyses. Gianluca Aguzzi, Lorenzo Bacchini, Martina Baiardi, Roberto Casadei, Angela Cortecchia, Davide Domini, Nicolas Farabegoli, Danilo Pianini, and Mirko Viroli present "A Demonstrator Toolchain for Self-organising Robot Teams", awarded two badges (available and reusable), which describes a demonstrator for deploying aggregate computing solutions in robot coordination scenarios, addressing key requirements such as localisation, program distribution, and hardware abstraction.

Furthermore, there is a work by Alessandro Aldini and Claudio Antares Mezzina on "Formalizing Errors in CCS with 3-Valued Logic", which adds to the process algebra CCS a term to characterise error configurations in concurrent systems, based on McCarthy's conjunction in three-valued logic. The work laid out includes an operational semantics, a sound axiomatization for bisimulation, an HML characterisation of the bisimulation, and a case study. A paper by Fabio Gadducci, Carlos Olarte, and Frank Valencia proposes "A Constraint Opinion Model" that generalises the standard DeGroot opinion model by representing agents' opinions and influences as soft constraints rather than single real values. The resulting model is now able to consider agents sharing partial information and preferences, engaging in discussions on multiple topics simultaneously, and representing opinions with different degrees of uncertainty, including imposing them. Moreover, Antonio Brogi, Roberto Casadei, Nicolas Farabegoli, Stefano Forti, and Mirko Viroli describe a "Declarative Deployment Planning for Green Pulverised Collective Computational Systems", an approach and toolchain based on simulation, coupled with a Prolog-based planner to guide the search for candidate deployments

of pulverised systems. The work shows the declarativity, modularity, scalability, and amenability of the setting for continuous reasoning of deployment alternatives. Finally, Carolyn Talcott develops "Dialects for the CoAP IoT Messaging Protocol", focusing on the Constrained Application messaging protocol, giving a specification of a resource-limited reactive attack model and its reachability analysis, demonstrating reactive attacks and their mitigation by dialecting.

We thank everyone involved in organising the conference. We are especially grateful to the authors for their valuable submissions, to the program committee and reviewers for their thoughtful evaluations that helped shape a strong technical program, and to the artefact committee for their crucial role in standardising and assessing the submitted tools.

We also thank the providers of the EasyChair Conference Management System, which was a great help in organising the submission and reviewing process and in the preparation of this proceedings volume.

April 2025

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Invited Contributions

Mechanized Type Soundness for Substructural Types using Iris (Invited Tutorial)

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Abstract. Substructural type systems are a good fit to enforce strong safety properties of higher-order, imperative and concurrent programs. Key examples are the Rust type system (which enforces the absence of undefined behavior and data races) and session types (which enforce correct usage of message-passing channels). Formally proving that these type systems "do their job" (i.e., they enjoy the *type soundness* property) is challenging. This challenge is exacerbated when considering combinations of language features ("feature interaction") and linking against "unsafe" libraries (which have to be verified manually), thus providing the desire for a scalable approach with support for machine-checked proofs.

In this tutorial I will give an introduction to the *logical approach* to type soundness, which is well-suited to prove soundness of substructural type systems. I will explain the theory of the logical approach in the context of a very simple language, which I then gradually scale up to a session-typed language. I will also demonstrate that the logical approach is well-suited for the mechanization of challenging type systems in the Rocq proof assistant using the Iris framework for concurrent separation logic.

Keywords. Substructural type systems · Type soundness · Separation logic · Rocq proof assistant

1 Introduction

A key property of a type system is type soundness (also called type safety), which asserts that "well-typed programs cannot go wrong". Introduced by Milner more than 4 decades ago [17], type soundness is among the most widely studied properties in programming languages.

Type soundness is a difficult property to prove for advanced substructural type systems which enforce correct usage of resources. For instance, the Rust language uses substructural types to enforce the absence of undefined behaviour due to memory errors (such use-after-free) and data races. Session types [6, 9] use substructural types to enforce correct usage of message-passing channels.

Type soundness becomes more challenging to prove when considering libraries implemented using "unsafe" language features (such as unchecked type casts). For

instance, Rust uses unsafe code for the implementation of efficient (concurrency) libraries for *e.g.*, atomic reference counting and mutexes. In session types it is desirable to treat message-passing channels as a library (implemented using efficient low-level unsafe code) rather than a built-in feature of the language. The correctness of these unsafe libraries is outside of the scope of the type system, and they thus need to be verified manually in a way that their soundness in combination with the rest of the type system is guaranteed.

The prevalent technique for proving type soundness is the syntactic approach of progress and preservation [21]. While widely used, the syntactic approach is not applicable to unsafe language features, and when applied to a substructural type system (especially in a proof assistant) it easily leads to an abundance of tedious low-level side-conditions (related to disjointness of resources).

In this invited tutorial I will discuss the logical approach to type soundness as an alternative technique that solves these problems [4, 20]. The logical approachis inspired by the old idea of semantic type soundness—as used in Milner's original paper on type soundness [17], and later extended by Appel, Ahmed, and their collaborators [1–3]—where types are given a semantic meaning via a logical relations model. The logical approach extends the semantic approach by employing a program logic to perform the heavy lifting. One does not reason in terms of the low-level operational semantics of the programming language, but uses the high-level abstractions and proof rules of the program logic. This provides a clear separation of concerns where the program logic encapsulates the semantics of orthogonal aspects such as resources (through concurrent separation logic [5, 18, 19]), generics and modules (through higher-order logic) and recursive structures (through step-indexing [1, 3]).

The program logic that we use is **Iris** [11–13, 15], a higher-order concurrent separation logic. An important feature of Iris is its Proof Mode [14, 16], a tactic language for separation logic embedded in the Rocq proof assistant. The logical approach using Iris in Rocq has been used successfully for among others a safety proof of Rust and some of its standard libraries [12] (in the RustBelt project), as well as session types [7, 8, 10] (in the Actris project). An extensive discussion of the logical approach and its applications is given in a recent journal paper [20].

2 Overview of the Tutorial

The tutorial consists of two parts:

1. In the first part, I will explain the theory of the logical approach to type soundness. I will do so in the context of a small simply-typed language with a substructural type system, which I gradually extend with more advanced features (polymorphism, existential types, recursive types, and references). I will also implement and verify session channels as an unsafe library.
2. In the second part, I will demonstrate how to use the Iris Proof Mode in the Rocq proof assistant. I will first show how to carry out basic proofs in separation logic and then apply the Iris Proof Mode to mechanize the type soundness proof developed in the first part of the tutorial.

Acknowledgments. I thank my coauthors on the Iris, RustBelt, and Actris projects. The author is supported, in part, by an ERC Consolidator Grant (project COCONUT, grant agreement no. 101171349), funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

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Towards Automated Analysis of Emerging Behaviour: Preliminary Insights and Research Directions (Invited Talk)

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Abstract. In several classes of collective systems, sophisticated dynamics may occur for the combined effect of the elementary actions of the individuals. For instance, foraging ants in a colony are able to collectively work out the most convenient path while carrying food back to their nest. Somehow similarly, unintended order may arise spontaneously among the involved parties of some economic systems.

Fascinating phenomena of this kind are studied in a variety of disciplines, from biology to physics, where the main challenge is to understand the relationship between the way individuals behave in isolation and what happens over time at a global level in the system.

This talk summarises a few proofs of concept on the automated analysis of different systems, such as stable matchings, flocks of birds, colonies of ants, and applauding audiences, and relevant emerging properties thereof. The underlying methodology follows an integrated approach to formal specification and analysis to leverage intuitive specification languages as well as existing mature verification technology and decision procedures for general-purpose applications.

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